

Unit – 4 : Session – 3 : SLO – 1 | Custom Higher-Order Functions

I. Fill in the Blanks

1. `lambda` — a lambda function in Python is a short anonymous function.
2. `function` — a higher-order function can take another function as an argument (or return one).

II. Multiple Choice

1. `list(filter(lambda x: x % 2 == 0, [1, 2, 3, 4]))` $\rightarrow$ `[2, 4]` — filter keeps only the even numbers.

III. Example — Custom Higher-Order Function

1. `def apply_twice(func, value): return func(func(value))`
2. `apply_twice(lambda x: x * 3, 2)` $\# \rightarrow 18$
3. Higher-order functions treat functions as first-class values: they can be passed, returned and stored like any other object.